| INCH-POUND |

MIL-F-3268E
17 July 1992
SUPERSEDING
MIL-F-3268D
11 February 1983

MILITARY SPECIFICATION

FITTINGS; PIPE, STEEL, SPECIAL (FOR TANKER UNLOADING INSTALLATIONS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers special fabricated steel pipe spool pieces and 90-degree elbows for use in tanker unloading operations.

* 1.2 Classification. Fittings are of the following types and sizes, as specified (see 6.2):

TYPE

Type A - Spool piece.
Type B - Spool piece and elbow.
Type C - Spool piece with tap-in.
Type D - Spool piece with tap-in and elbow.
Type E - Elbow.

SIZE

Size 6 - 6-inch nominal pipe size (NPS).
Size 8 - 8-inch NPS.
Size 12 - 12-inch NPS.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, 621 Pleasant Valley Road, Port Hueneme, CA 93043-4300, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 4730

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-R-2765 - Rubber Sheet, Strip, Extruded, and Molded Shapes, Synthetic, Oil Resistant.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN PETROLEUM INSTITUTE (API)

API STD 1104 - Welding of Pipelines and Related Facilities.

(Application for copies should be addressed to the American Petroleum Institute, 1220 "L" Street, NW, Washington, DC 20005.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings.
ASME B16.9 - Factory-Made Wrought Steel Buttwelding Fittings.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

ASTM

ASTM A 53 - Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless.
ASTM A 105/A105M - Forgings, Carbon Steel, for Piping Components.
ASTM A 234/A234M - Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a fitting shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.2.1 Steel. Buttwelded type fittings shall conform to ASME B 16.9, standard weight, material to ASTM A 234/A234M. Standard weight steel pipe shall conform to ASTM A 53, type F or G, grade A. Forged steel pipe flanges and flanged fittings shall conform to ASME B16.5, class 150, material to ASTM A 105/A105M.

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3.2.2 Gasket material. Gaskets shall be fabricated from material conforming to MIL-R-2765, not less than 0.0625-inch thick. The material shall be tested in accordance with 4.5.1.

3.3 Design. The fitting(s) shall be fabricated as shown on figure 1. Each fabricated fitting shall withstand an internal hydrostatic pressure of 350 pounds per square inch (psi).

3.3.1 Spool piece. The spool piece shall include a standard weight steel pipe, approximately 2 feet long, with a slip-on flange welded to each end. Two lifting ears of plate steel, spaced 180 degrees apart as shown in figure 1, shall be welded longitudinally to the pipe body midway between the flange faces, and approximately in line with the midpoints between the pairs of bolt holes in each flange. The lifting ears shall be welded to the pipe with a 0.375-inch fillet weld all around.

3.3.2 Spool piece tap-in. When a spool piece with tap-in is required (see 1.2 and 6.2), the spool piece shall have two 1.25-inch outside diameter by 0.25-inch thick bosses, welded 5.5 inches from one end, as shown on figure 1. The two bosses shall be spaced 180 degrees apart. A hole shall be drilled through the bosses and the pipe, and tapped with 0.25-inch American standard taper pipe thread (NPT). Fittings with 0.25-inch NPT threaded connections and contoured at the base may be used to provide threaded outlets in lieu of bosses, upon approval of the contracting officer. Each threaded outlet shall be fitted with a 0.25-inch steel plug.

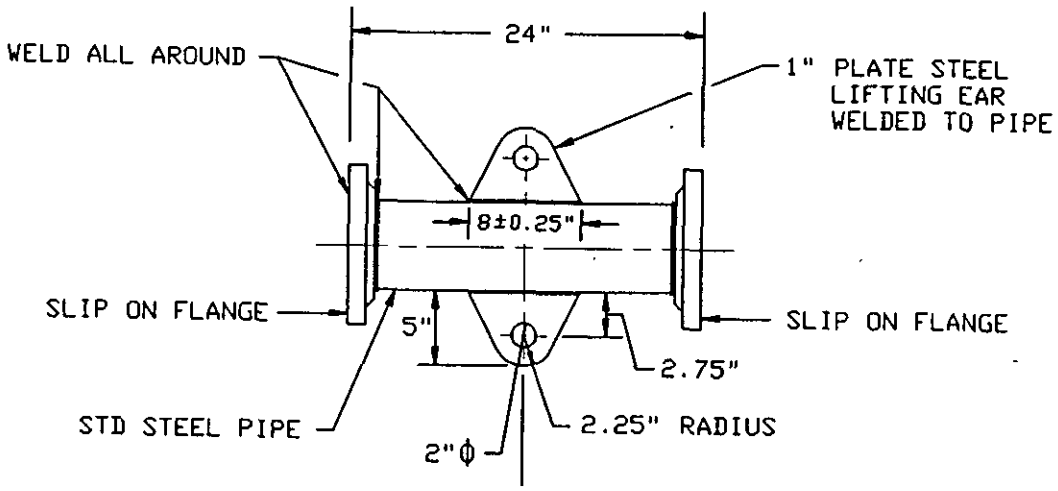
3.3.3 Elbow. The elbow shall include a welded neck flange; a standard weight, black steel, 90-degree long radius buttwelded elbow; a buttwelded lap-joint stud end; and a lap-joint flange loosely fitted to the stub end. A lifting ear of plate steel shall be welded longitudinally to the long radius of the 90-degree elbow, approximately midway between the ends, by a 0.375-inch fillet weld all around.

3.3.4 Gaskets. Unless otherwise specified (see 6.2), gaskets shall be furnished for all flanges, except when spool piece (with or without tap-in) and elbow are required. When the spool piece and elbow are required and gaskets are to be furnished, one gasket for the elbow end, one gasket for the spool piece end, and one gasket between the mating ends of the spool piece and elbow shall be furnished. When the spool piece and elbow are required, and gaskets are not to be furnished, one gasket between the mating ends of the spool piece and elbow shall be furnished. The gaskets shall be pre-cut and shall have the following inside and outside diameters for the specified NPS (see table I):

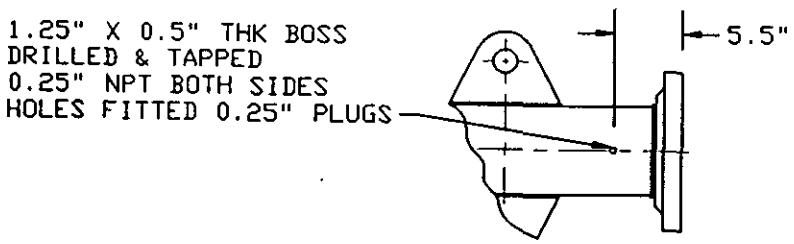
TABLE I. Gasket size and tolerance.

NPS	Inside diameter (inches)	Outside diameter (inches)	Tolerance (inches)
6	6	8.625	-0 +0.0625
8	8	10.875	-0 +0.0625
12	12	16	-0 +0.0625

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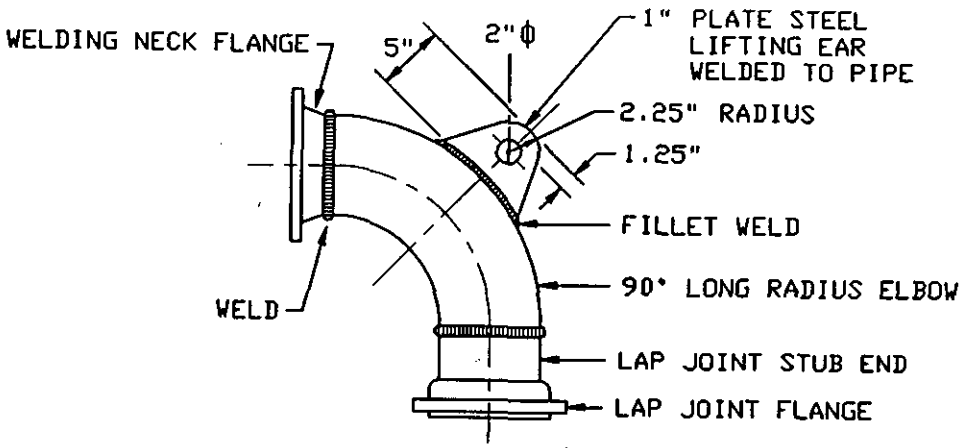
A. SPOOL PIECE



Note:

1. All other dimensions same as figure A.

B. SPOOL PIECE WITH TAP-IN



C. ELBOW

FIGURE 1. Fittings, pipe, steel, special (for tanker unloading installations).

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3.4 Identical items. All fittings of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.5 Identification marking. The manufacturer's name, initials, or trademark shall be legibly cast or die-stamped on each fitting.

3.6 Cleaning, treatment, and painting. Exterior surfaces of fittings, except flange faces, shall be cleaned, treated, and painted in accordance with MIL-T-704, type B. The color of the finish coat shall be as specified (see 6.2).

3.7 Workmanship.

3.7.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.7.2 Welding. Surface to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with API STD 1104. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one fitting assembly when a first article is required (see 3.1 and 6.2). This inspection shall include the examination of 4.4, and the tests of 4.5.1, 4.5.2, and 4.5.3. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests 4.5.2 and 4.5.3, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

* 4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one fitting assembly. All fittings offered for delivery at one time shall be considered a lot for the purpose of inspection. Guidance for inspection level and Acceptable Quality Level (AQL) is provided in 6.5.

4.4 Examination. Each fitting shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests.

4.5.1 Gasket material test. The gasket material shall be tested in accordance with MIL-R-2765.

4.5.2 Hydrostatic test. Each sample fitting shall be subjected to an internal hydrostatic pressure of 350 psi for 15 minutes. Any evidence of leakage shall be considered a defect and the fitting shall be rejected.

4.5.3 Welding test. Radiographic inspection procedures and tests shall be conducted on the buttwelds of elbow fittings in accordance with API STD 1104 to determine if the welds conform to the requirements of 3.7.2.

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4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-V-3 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fittings are intended for use with submerged loading line installations in conjunction with tanker unloading operations of liquid fuels. The spool piece is used for hoisting and securing the hose which connects the submerged pipeline to the ship receiving or discharging fuel.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size of fitting required (see 1.2 and 3.3.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required for inspection and approval (see 3.1 and 4.2.1).
e. When a gasket is not required (see 3.3.4).
f. Color of finish coat required (see 3.6).
g. Level of preservation and packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

* 6.5 Sampling procedures.

* 6.5.1 Sampling for examination. Recommended inspection level is II and AQL is 4.0 (see 4.3).

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* 6.5.2 Sampling for tests. Recommended inspection level is S-3 and AQL is 1.5 (see 4.3).

* 6.6 Part or identifying number (PIN). The PIN to be used for fittings applied to this specification is created as follows:

	M3268	XX
Military specification designation		
Type and size (NPS) code (see table II)		

* 6.6.1 Type and size of fitting. The type and size of fitting are identified by a two digit number (see table II).

TABLE II. Code number to type and size.

Type of fitting	Size 6	Size 8	Size 12
Type A	01	02	03
Type B	04	05	06
Type C	07	08	09
Type D	10	11	12
Type E	13	14	15

* 6.7 Subject term (key word) listing.

Elbow
Gasket
Spool piece

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - YD
Air Force - 99

Review Activities:
Air Force - 82
DLA - CS

User Activity:
Army - CE

Preparing Activity:
Navy - YD

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-3268E	2. DOCUMENT DATE (YYMMDD) 920717

3. DOCUMENT TITLE
FITTINGS; PIPE, STEEL, SPECIAL (FOR TANKER UNLOADING INSTALLATIONS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Miss Anne L. Lyons	805-982-5761	551-5761
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NCBC CODE 1564A, 621 PLEASANT VALLEY ROAD PORT HUENEME, CA 93043-4300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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